

Does the Macro Cycle Improve a Capital-Distress Scorecard for Brazilian Banks?

Credit Risk modelling

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The macro overlay adds +0.0067 AUC on paper and nothing that survives a clustered test

EXECUTIVE SUMMARY

+0.0067

out-of-time AUC gain from macro, random forest

0.152

p-value once we resample institutions, not rows

6.1%

of the forest's importance carried by the three macro columns

6

distinct SELIC values in the test year

01 We built the course scorecard on a prudential-conglomerate panel of 229 institutions. Four WoE-transformed CAMELS ratios give an out-of-time AUC of **0.8652** with no train-to-test gap, and a random forest on the same inputs ties it at 0.8620.

02 Then we merged three Banco Central macro series, SELIC, IPCA and IBC-Br, onto every row by month and refitted the scorecard and the forest with the three columns added. Nothing else in the pipeline changes.

03 The scorecard doesn't move (0.8652 to 0.8646). The forest gains **+0.0067** (0.8620 to 0.8688). Resampling rows calls that significant. Resampling the 23 distressed institutions gives $p = \mathbf{0.152}$, the macro columns carry 6.1% of the forest's importance, and SELIC takes only six values in the test year.

04 PSI caught a real break in the data. It stays below 0.08 for eight years, then reaches **3.7566** in 2025 when CMN Resolution 4.966 removes the source accounts of two of our four variables.

The Through-the-Cycle WoE scorecard stays the primary model. The macro overlay is not adopted. The random forest, explained with SHAP, stays as the challenger.

Does the Macro Cycle Improve a Capital-Distress Scorecard for Brazilian Banks?

WHAT WE ARE GOING TO DO

Our question

“Does adding the macro cycle, SELIC, IPCA and IBC-Br, to a CAMELS scorecard improve out-of-time prediction of a capital breach within twelve months?”

1

Build the baseline the course way

A prudential-conglomerate panel, four WoE-transformed CAMELS ratios and a logistic scorecard, fitted on train and judged once on an untouched 2023.

2

Merge the macro cycle onto every row

Three Banco Central SGS series joined by month. The scorecard and a random forest are refitted with and without the three columns, on identical rows.

3

Decide with a test that respects the sample

A paired bootstrap clustered by institution. A gain counts only if it survives resampling the 23 conglomerates that go bad in the test year.

- 01 The three macro series are the only inputs the course pipeline doesn't already hold.
- 02 Same rows, same four ratios and same test year for every model, with and without macro.
- 03 We report the result either way, including the one we didn't want.

■ If the overlay doesn't beat the scorecard on a year the models never saw, we keep the scorecard.

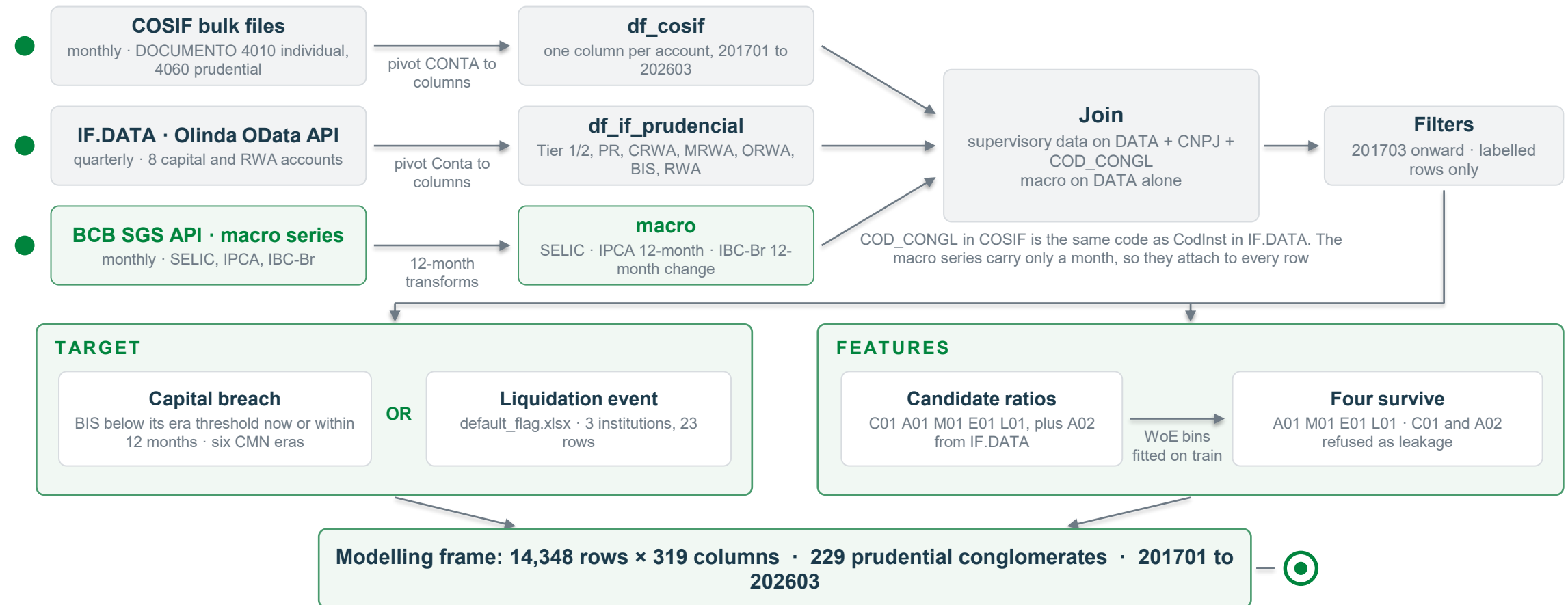
01

The baseline

The course scorecard on a prudential-conglomerate panel: two supervisory sources, a capital-distress label, four CAMELS ratios and one logistic regression. The overlay is judged against this.

Two supervisory sources build the panel, and a third source, the macro series, is what we add

THE BASELINE



■ Everything above the macro row is the course pipeline. The macro row joins on nothing but the month.

Modelling the conglomerate, not the bank, takes labelled rows from 47.3% to 95.3%

THE BASELINE

Our label is a breach of a consolidated regulatory capital ratio, and that ratio is reported for the **prudential conglomerate**, not for the individual bank. CAMELS ratios built at bank level describe one entity while the label describes another. The cost is that institutions outside a prudential conglomerate leave the panel. We accept it, because a smaller panel that is specified correctly is worth more than a larger one that is not.

	Individual-bank panel	Prudential-conglomerate panel	What it means
Rows carrying a usable label	47.3%	95.3%	Most of the bank-level panel could not be labelled at all
Out-of-time bad institution-months	31	207	31 events cannot support a model comparison; 207 can
Bad rate, once a missing ratio is read as unknown rather than as a breach	16.44%	9.10%	The 16.44% was mostly missing data, not distress

Three corrections this forced:

	Correction	Before	After
V1	A missing capital ratio is labelled unknown , not breach	bad rate 16.44%	9.10%
S1	C01 (the Basel ratio itself) and A02 are not offered to the model	out-of-time AUC 0.899	0.864 , and the higher figure was the leak
S3	train and out-of-time split moved inside the COSIF 1.0 accounting regime	100 bad events	207

■ Each of these three corrections lowered a number we could have reported.

Our target is capital distress within twelve months, not liquidation

THE BASELINE

target = 1 if the conglomerate is below its regulatory capital threshold in the current month **or in any of the following twelve months**, or carries an explicit liquidation flag from the instructor's default register. A **missing** capital ratio is labelled unknown and the row leaves the modelling sample.

Basel ratio threshold, applied to BIS in percent (CMN 4553 → 4783)	≤ 201512	≤ 202003	≤ 202103	≤ 202109	≤ 202203	thereafter
	11.0	10.5	9.25	9.625	10.0	10.5

9.25 is the COVID-19 capital-conservation relief; the schedule returns to 10.5 from 202204.

Sample	Periods	Rows	Institutions	Bad institution-months	Bad institutions	Bad rate
train	201703-202212	7,975	167	680	36	8.53%
out-of-time	202301-202312	1,737	157	207	23	11.92%
monitoring	202401-202603	3,954	n/a	362	31	9.16%

Read the out-of-time row carefully. It holds **207 bad institution-months from only 23 conglomerates**, and 16 of those 23 appear in all twelve months. The number of distinct institutions, not the row count, is what governs every significance statement we make, including the one about the overlay.

648 of the 1,263 bad rows (51%) are already in breach at the observation date, and only 3 institutions / 23 rows carry an explicit liquidation flag.

We keep four CAMELS ratios and drop the two with the highest information value

THE BASELINE

	What it actually measures	IV (train)	Status
A01	Impaired loans, risk levels D to H, over the classified credit portfolio	0.4193	in the model
M01	Net non-operating result over total assets , not a net interest margin	0.2043	in the model
E01	Net result over equity	0.9664	in the model
L01	Cash and equivalents over deposits , not a current ratio	0.1912	in the model
C01	The Basel ratio itself, and the label is a threshold on this same series	1.1930	excluded (circular)
A02	Credit RWA over total assets, same IF.DATA source as the label, and it ends at 202306	0.5480	excluded (leakage)

We **fit the direction of each variable on train rather than assuming it**. Two of the four come out opposite to the course's assumption. A large non-operating result relative to assets (M01) is a distress signal, not a healthy one: it is asset disposals and one-off gains booked to shore up a bad year. Cash far exceeding deposits (L01) marks a thin, wholesale-funded institution rather than a liquid one. Under the assumed direction, optimal binning finds **zero** usable splits for L01 and we would have dropped it.

■ A high information value isn't evidence a variable belongs in the model. Here it was evidence of the opposite.

Four coefficients with the expected signs give the baseline an out-of-time AUC of 0.8652

THE BASELINE

	coef	std err	z	odds ratio	dy/dx
const	2.3849	0.0471	50.58	10.8577	n/a
A01_woe	0.6864	0.0694	9.88	1.9866	0.0447
M01_woe	0.8025	0.0906	8.86	2.2310	0.0523
E01_woe	0.8884	0.0394	22.57	2.4312	0.0579
L01_woe	0.5768	0.1271	4.54	1.7802	0.0376

7,975
training rows

0.1924
pseudo R²

0.8652
out-of-time AUC

All four coefficients $p < 0.001$. Train AUC 0.8138, out-of-time 0.8652, Gini 0.7304, KS 0.5838.

Standardised contribution to the score:

E01 39.3% · A01 23.0% · L01 19.7% · M01 18.0%

The model predicts the probability of being **good**, so a positive coefficient means a higher Weight of Evidence raises the chance an institution gets through the next twelve months without a capital breach. All four are positive. The sign check passes, and no ratio argues against its economic reading.

Read as odds ratios: a one-unit rise in E01_woe multiplies the odds of being good by **2.43**, in M01_woe by 2.23, in A01_woe by 1.99 and in L01_woe by 1.78. Earnings carry the model. Liquidity contributes least, with the smallest coefficient and the lowest z.

What the baseline cannot do

Every input is a ratio the institution reports about itself, so the table gives the same probability for the same balance sheet in every month of the panel. It has no way to know that the policy rate went from 2% to nearly 14% during the training window. That is the gap the overlay is meant to fill.

This is the model the overlay has to beat, and its whole specification fits in one table.

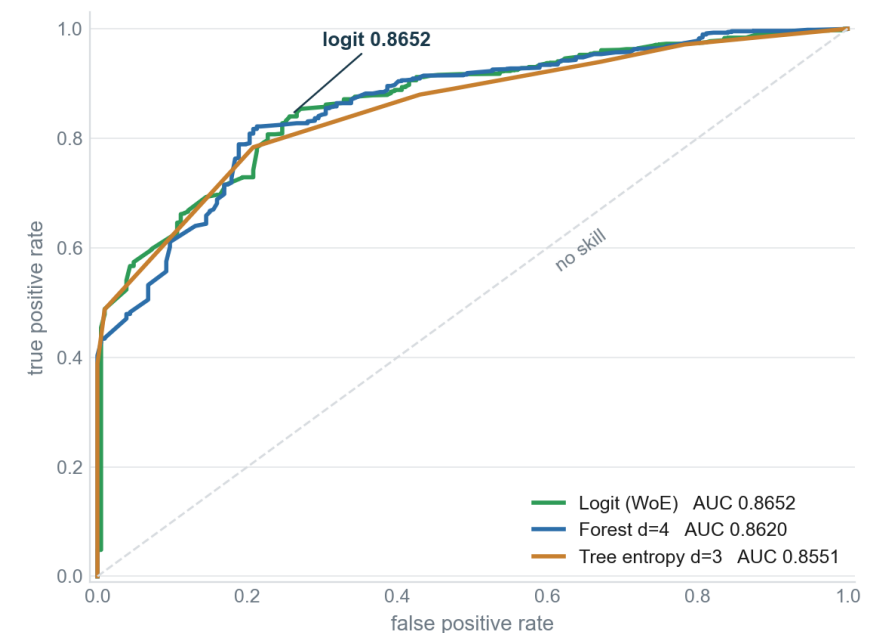
A tree and a forest tie the scorecard on the same inputs, so the forest carries the overlay

THE BASELINE

Model	train	oot	mr	oot Gini	oot KS
Logit (WoE)	0.8138	0.8652	0.6850	0.7304	0.5838
Tree entropy d=3	0.8000	0.8551	0.6742	0.7102	0.5759
Forest d=4	0.8339	0.8620	0.6902	0.7241	0.6093
Forest d=8	0.8632	0.8438	0.6650	0.6877	0.5298
Forest, unrestricted	0.8646	0.8325	0.6653	0.6650	0.5271
HistGB, raw ratios d=6	0.9914	0.8341	0.6201	n/a	n/a

The forest sits **0.0031** below the scorecard out-of-time, with a clustered p-value of 0.724. More depth buys training fit and loses it on the test year: the unrestricted forest reaches 0.8646 on train and 0.8325 out-of-time, and gradient boosting on raw ratios 0.9914 against 0.8341. WoE binning already fitted a monotone step to each ratio, so four inputs leave a tree little to find.

Why the forest is the second vehicle for the overlay: it takes SELIC, IPCA and IBC-Br as raw levels and can split on a macro value **inside** a ratio branch, so a rate spike can matter more for a thinly funded institution. The scorecard can only add the three columns. We run both.



■ The forest ties the scorecard before macro is added, so any macro gain in it isn't the forest catching up.

02

The overlay

Why the macro cycle should matter, which three series we merge, and how we test whether they add anything on a year the models never saw.

The course lists economic conditions as a driver of default, yet the scorecard can't see them

THE OVERLAY

Session 2 lists the borrower's financial condition, industry sector and **economic conditions** among the drivers of credit risk. Our four CAMELS ratios cover the first two and nothing else. Each is an internal balance-sheet ratio, so the scorecard returns the same probability for the same balance sheet whether SELIC stands at 1.90 or at 13.65.

That makes it a **Through-the-Cycle** score. A **Point-in-Time** score moves with the cycle, and a supervisor watching Brazilian banks through 2022 and 2023, with the policy rate rising from 2% to nearly 14%, would want one. The training window covers that whole cycle, so a model has something to learn from.

The overlay is the simplest version of the idea: the same four ratios, plus three monthly series as extra columns, in the scorecard and in the forest.

1.90 to 13.65

SELIC range across the training window, %
p.a.

none

macro inputs anywhere in the course
pipeline, Sessions 1 to 7

THE OVERLAY · THE SCORECARD WITH THREE MACRO COLUMNS

$$\text{logit}(P_{it}) = \beta_0 + \sum_{k=1}^4 \beta_k x_{k,it}^{\text{woe}} + \gamma_1 \text{SELIC}_t + \gamma_2 \text{IPCA}_t^{12m} + \gamma_3 \text{IBC}_t^{\text{yoy}}$$

What we expect each series to carry

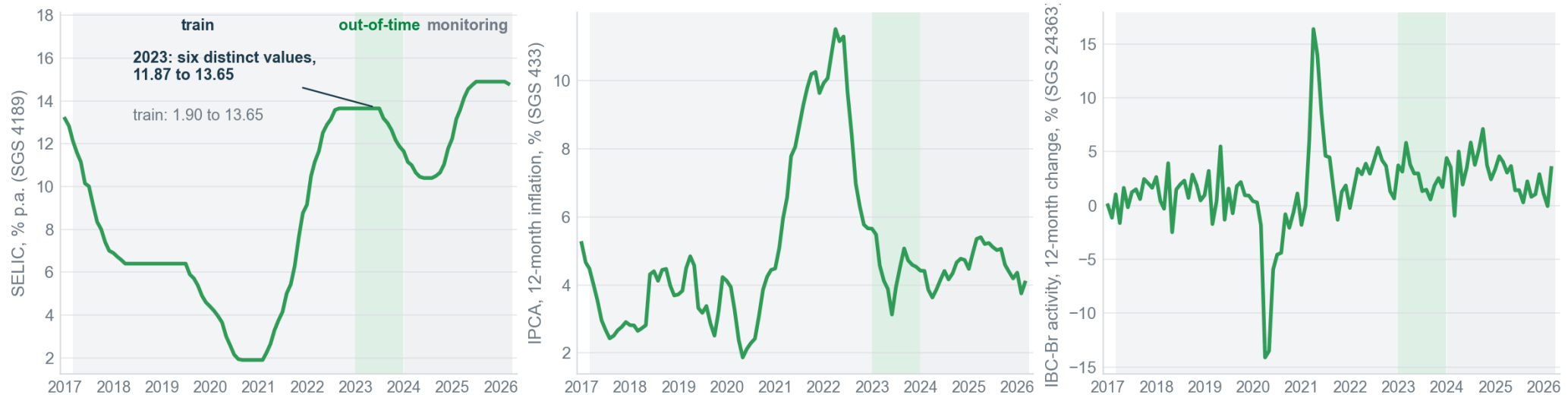
Series	Mechanism
SELIC	Funding cost and the value of fixed-rate books; a spike hurts thinly funded institutions first
IPCA, 12-month	Erodes real earnings and real capital; raises borrower default
IBC-Br, 12-month change	Activity growth lowers credit losses and lifts earnings

Same balance sheet, different month: a Through-the-Cycle score gives the same PD, a Point-in-Time score doesn't. The overlay is what separates the two, and whether it earns its place is an empirical question.

■ Adding the cycle is the smallest change that could turn a Through-the-Cycle scorecard into a Point-in-Time one.

Three monthly Banco Central series join on the month alone, with 100% coverage of the panel

THE OVERLAY



Series	SGS code	Transform we use	Coverage on the panel
SELIC, policy rate, % p.a.	4189, monthly	level	126 months, 100%
IPCA, consumer inflation	433, monthly	12-month sum	126 months, 100%
IBC-Br, activity index	24363, monthly	12-month change, %	126 months, 100%

A trap worth recording

SGS 432 is the daily SELIC. An unbounded request returns an HTTP 406 error object, not a series, and a fetch that swallows the error trains a model with no SELIC column while the write-up goes on crediting SELIC. We use 4189, the monthly series, and the fetch raises on anything that is not a series.

The merge is trivial, and that is the point: the series are public, monthly, and nothing in the course uses them.

The three macro columns are the only difference between the four models we compare

THE OVERLAY

Same inputs

- Every model gets the identical four WoE columns
- The macro variants add SELIC, IPCA_12m and IBC_yoy as raw levels
- No model sees anything else

Same rows

- Fitted on the identical 7,975 labelled training rows
- Scored on the identical 1,737 out-of-time rows
- Macro joined by month, 100% coverage

Two families, four models

- Logit WoE and Logit WoE + macro
- Forest d=4 and Forest d=4 + macro, 500 trees, class-balanced
- Same seed, same depth, same leaf size

One test

- Paired stratified bootstrap, B = 4,000
- Row-level shown, clustered by institution decides
- Nothing tuned on the test window

train · 201703-202212

7,975 rows · 680 bad institution-months · fitted here

out-of-time · 2023

1,737 rows · 207 bad from 23 · judged here

monitoring · 202401-202603

3,954 rows · 362 bad · never used to decide

Two comparisons decide the question: **Forest against Forest + macro**, and the scorecard against Forest + macro. A row-level bootstrap treats each of the 1,737 institution-months as independent. The clustered version resamples whole institutions, which is the unit the 207 bad rows actually come in.

- 01 A gain counts only if the interval clustered by institution excludes zero. The row-level interval is shown to make the difference visible.
- 02 Nothing is fitted, screened or tuned on 2023, so the test year is used exactly once.

■ If the overlay helps, it has to show up here, and it has to survive resampling 23 institutions.

03

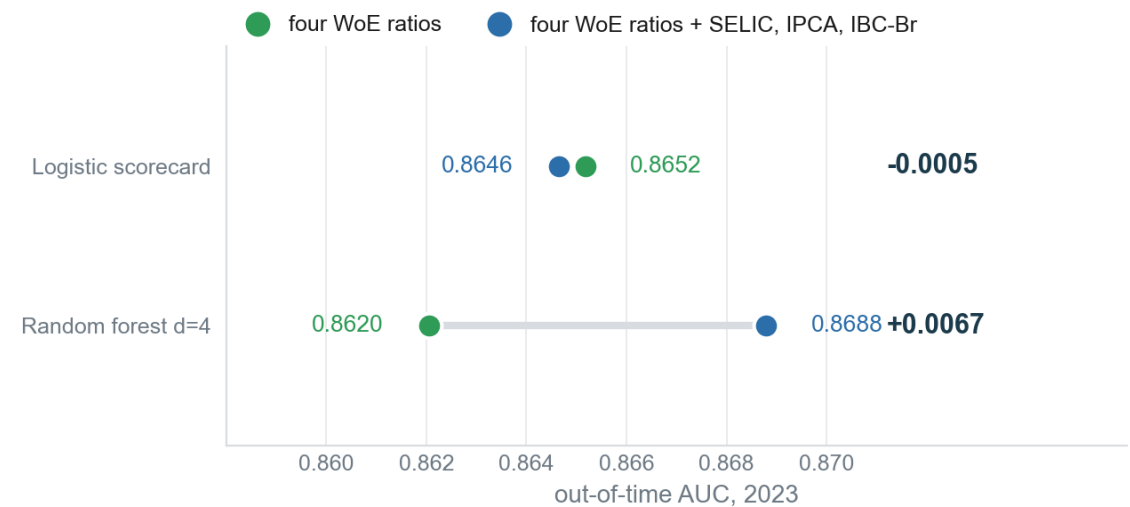
Results

What the overlay buys at face value, what is left of it once we resample institutions, and why the forest hardly uses the macro columns.

Macro moves the scorecard by -0.0006 and the forest by $+0.0067$ on the test year

RESULTS

Model	train AUC	out-of-time AUC	mr AUC
Logit WoE (baseline)	0.8138	0.8652	0.6850
Logit WoE + macro	0.8137	0.8646	0.6882
Forest WoE d=4	0.8339	0.8620	0.6902
Forest WoE + macro d=4	0.8361	0.8688	0.6853



The scorecard can't use the columns the way we hoped. It adds them linearly, so a one-point rise in SELIC shifts every institution's score by the same amount in that month. That re-levels months and reorders nobody within one, and AUC is a ranking measure.

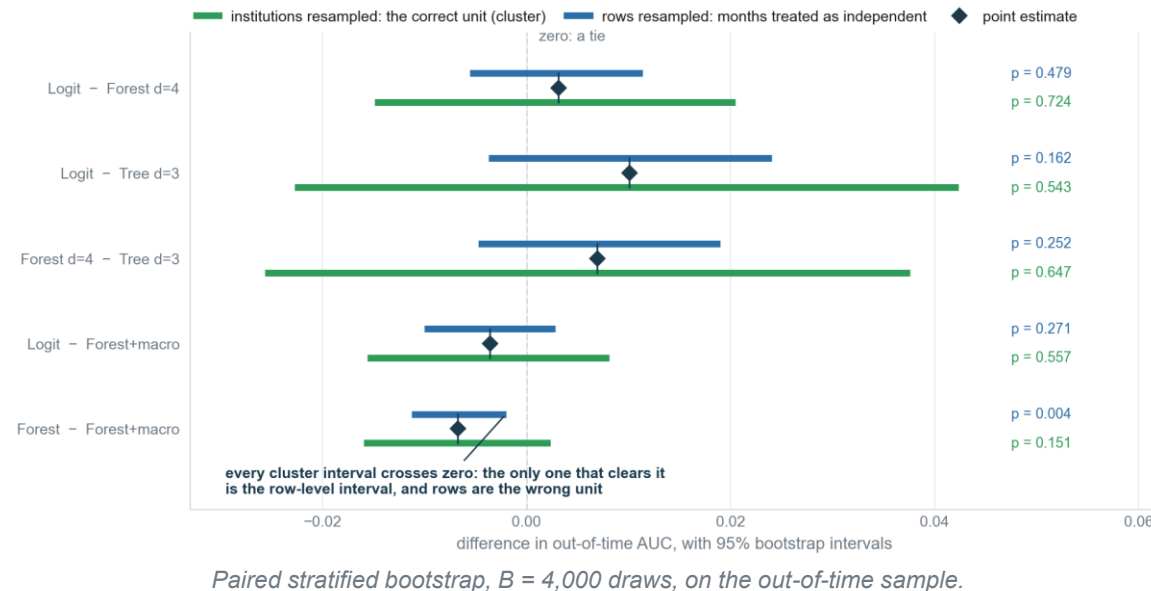
The forest can split on SELIC inside an E01 branch, so it can reorder institutions. That is where **+0.0067** on the test year comes from, and the forest gives 0.0049 of it back in the monitoring window.

At face value, Forest + macro is the best model in this deck. Two things stop us from writing that down: the gain is measured on 23 distressed institutions, and we have not yet asked how much of it is macroeconomics.

■ On the point estimates the overlay helps the forest and not the scorecard. Whether **+0.0067** is real is a question about 23 institutions.

Resample institutions instead of rows and the macro gain is no longer distinguishable from zero

RESULTS



Comparison	ΔAUC	row p	cluster 95% CI	cluster p
Forest - Forest + macro	-0.0067	0.0035	[-0.0159, +0.0024]	0.152
Logit - Forest + macro	-0.0036	0.271	[-0.0156, +0.0081]	0.557
Logit - Forest d=4	+0.0031	0.480	[-0.0149, +0.0205]	0.724
Logit - Tree d=3	+0.0101	0.162	[-0.0227, +0.0424]	0.543

Row by row, the gain looks real: $p = 0.0035$. But the 207 bad rows are **23 conglomerates**, 16 of them present in all twelve months, so a month is not an independent event. Resample institutions and the interval is [-0.0159, +0.0024], $p = 0.152$.

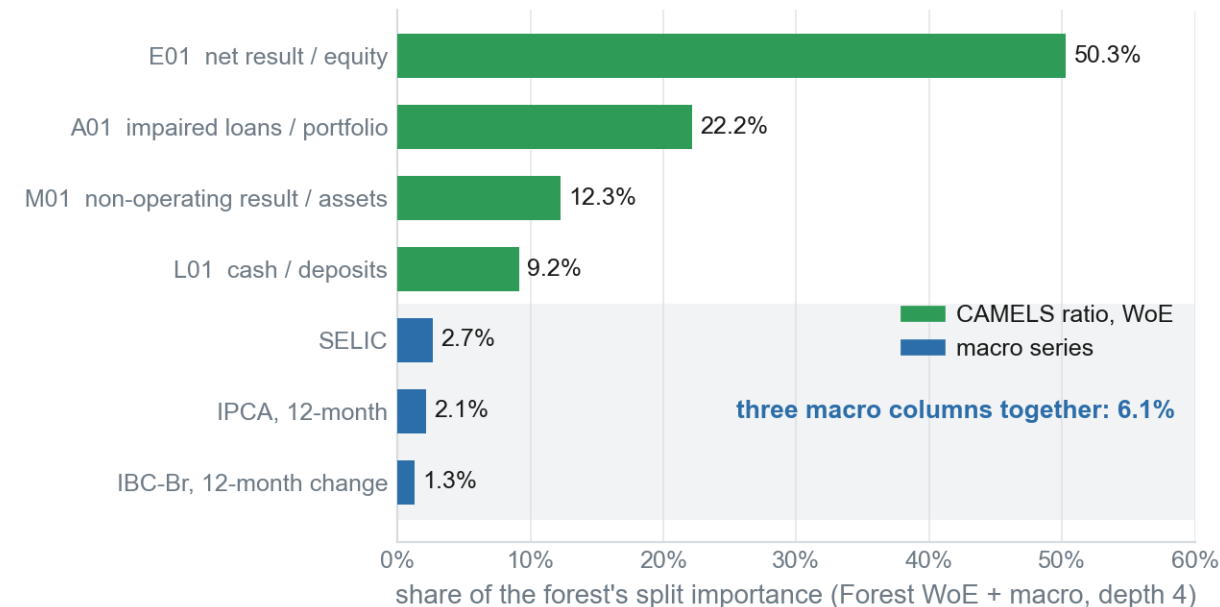
Every other comparison was already a tie. The overlay was the one result on the chart, and the unit of resampling was what made it look like one.

The discrimination itself is not an artefact of repeated rows. Collapse to one row per institution and out-of-time AUC is 0.8965 (train 0.8151).

■ **23 institutions is the real sample size, and it can't resolve a gain of 0.0067.**

The three macro columns carry 6.1% of the forest's importance, and 2023 holds six SELIC values

RESULTS



6.1%

share of split importance in the three macro columns

6

distinct SELIC values in the test year, 11.87 to 13.65

Inside the forest, E01 alone carries **50.3%** of the split importance and the four ratios **93.9%**. SELIC gets 2.7%, IPCA 2.1% and IBC-Br 1.3%. The forest is not ignoring the cycle; it is using the balance sheet first and the cycle as a footnote.

The test year gives it little else to do. Train spans SELIC from **1.90 to 13.65**, a full easing and tightening cycle. The 2023 window sits on the plateau at the top of it, with **six distinct values**. A column that is constant within a month and nearly constant across the year can only re-sort institutions between months.

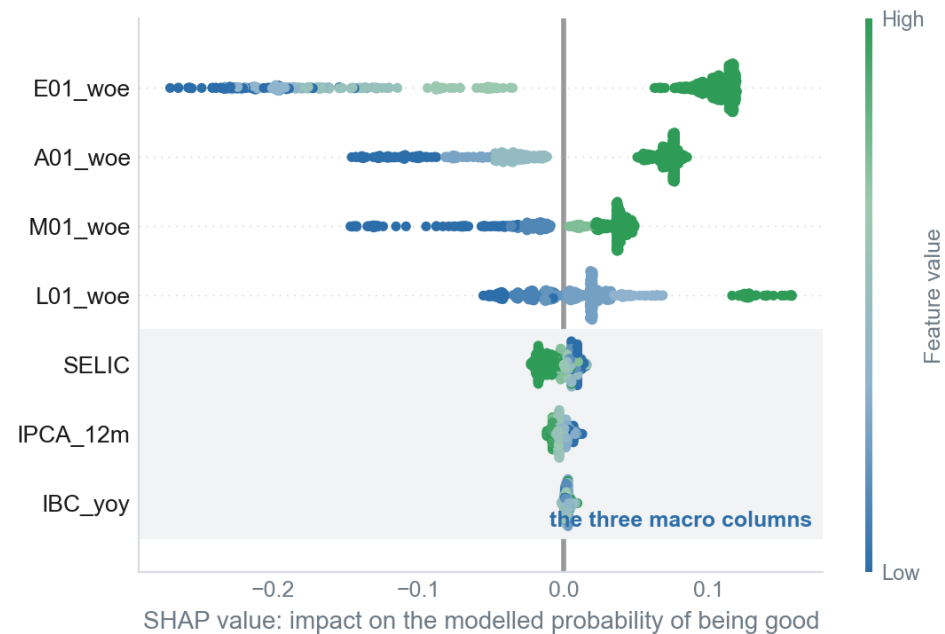
What this result is, and what it isn't

It is a null on one test year at one point in the cycle, from three additive columns. It is not evidence that the cycle doesn't drive capital distress. The natural next test is an interaction between the macro state and an institution's own liquidity, on a panel that spans more than one rate cycle.

We can't say the cycle doesn't matter for Brazilian banks. We can say this test year can't show that it does.

SHAP puts the three macro columns at the bottom of the forest's ranking

RESULTS



Mean SHAP , out-of-time sample	value
E01_woe	0.1223
A01_woe	0.0640
M01_woe	0.0367
L01_woe	0.0293
SELIC	0.0093
IPCA_12m	0.0043
IBC_yoy	0.0023

The macro columns are not noise to the forest. High SELIC pushes the modelled probability of being good **down**, as we expected, and the effect is small and one-directional. It is also a tenth of what E01 does and a third of what L01, the weakest ratio, does.

This is what makes the forest usable as a challenger: its ranking can be read, and it agrees with the scorecard on the order of the four ratios.

■ Explainable, and explained: the forest reads the cycle as a footnote to the balance sheet.

04

Rating, monitoring, verdict

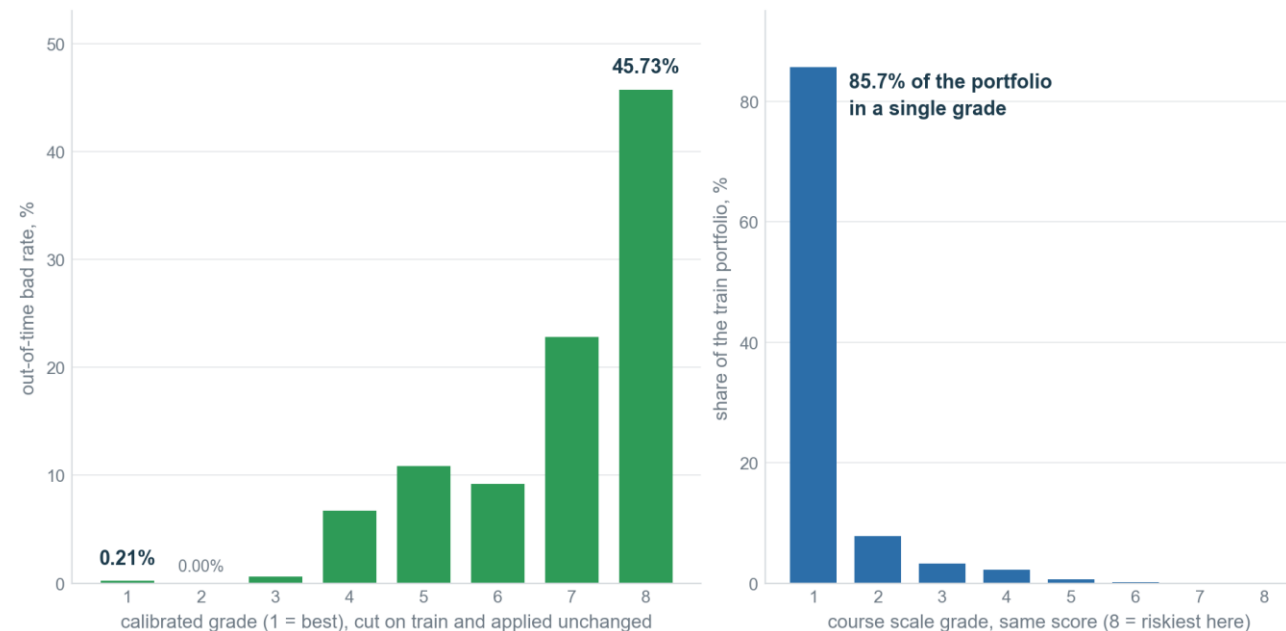
An eight-grade scale cut once on train, a PSI alarm that caught a real break in the data upstream, and the answer to the question we started with.

The course scale puts 85.73% of the portfolio in one grade, so we cut our own

RATING, MONITORING, VERDICT

Applied to our training sample, the documented cut-offs put **85.73%** of the portfolio into grade 1 at a 4.68% bad rate, while grade 8 holds 0.05% at a 100% bad rate. Grade 1 is therefore the safest and grade 8 the riskiest, which is **inverted** relative to the documented “8 = best”. We report both problems rather than quietly correcting them.

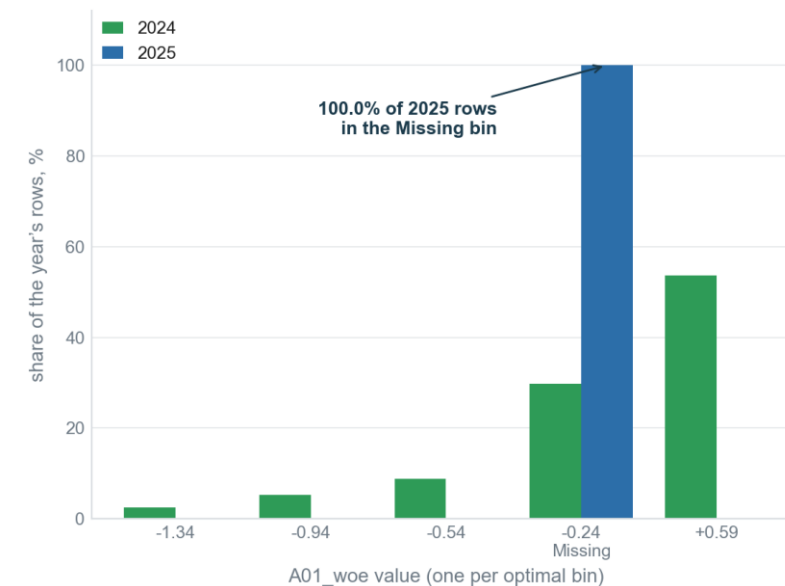
Our replacement is a quantile scale with **1 = best**, cut on train only and applied unchanged to the out-of-time and monitoring windows. It ranks correctly out-of-time, grade 1 at **0.21%** against grade 8 at **45.73%**, but it under-predicts the level: **9.66% predicted against 11.92% observed**.



■ A scale cut once on train and never touched again separates 0.21% from 45.73% out-of-time.

PSI flagged the 2025 chart-of-accounts change in the year it happened

RATING, MONITORING, VERDICT



Aggregate PSI against train: out-of-time **0.0581** (stable), monitoring window **0.5138** (alarm). By year the series is flat at 0.0154, 0.0043, 0.0139, 0.0039, 0.0111, 0.0097, 0.0581 and 0.0753, then jumps to **3.7566** in 2025 and **3.7084** in 2026.

The cause is diagnosed, not guessed. From 202501 the CMN Resolution 4.966 chart of accounts (COSIF 1.5) removed the source accounts for A01 and M01, and the crosswalk has no counterpart for them, so **both variables are 100% missing**. A01_woe spreads 53.7 / 29.7 / 8.8 / 5.2 / 2.5% across its bins in 2024 and is **100.0% the Missing bin** in 2025. The model receives a constant for two of its four inputs, which is why monitoring AUC falls to about 0.68 for **every** model, with or without macro.

■ **Monitoring would have suspended the score before anyone took a 2025 decision on it.**

The Through-the-Cycle scorecard stays, and the overlay is recorded as tested, not adopted

RATING, MONITORING, VERDICT

What we set out to do	What happened
Build the baseline the course way, on a conglomerate panel	Done. Out-of-time AUC 0.8652, no train-to-test gap, four coefficients with the expected signs, all $p < 0.001$
Merge SELIC, IPCA and IBC-Br onto every row and refit	Done. 100% coverage. The scorecard moves -0.0006 , the forest $+0.0067$ out-of-time
Decide with a bootstrap clustered by institution	Done, and the gain does not survive it. $p = 0.152$, 6.1% of importance, six SELIC values in the test year. Overlay tested, not adopted

Between the scorecard and the forest, Session 7 slide 28 decides

Dimension	Logistic scorecard	Random forest	Winner
Interpretability	Four signed coefficients, odds ratios and marginal effects	SHAP after the fact; single trees are unstable	logit
Accuracy on non-linear patterns	out-of-time AUC 0.8652	0.8620, $\Delta AUC +0.0031$, cluster $p = 0.724$	tie
Overfitting risk	train 0.8138 against out-of-time 0.8652, no gap	train up to 0.9914 against 0.8341 on raw ratios	logit
Governance and audit fit	Standard scorecard; WoE bins auditable per variable	Workable with an explainability layer; heavier to document	logit
Typical use case	Regulatory capital, pricing, the rating scale	Challenger, early warning, variable discovery	both

■ **Primary model: the WoE logistic scorecard. Challenger: the random forest with SHAP. Macro overlay: tested, not adopted.**

Appendix A · The full change register

APPENDIX

	Notebook	Change	Severity	Effect on the numbers
N1	01	Model at prudential-conglomerate level only, not a stack of three levels	High	Rows usable for fitting 9,141 → 13,883; oot bad events 31 → 207
V1	02	Missing capital ratio → unknown, not breach	High	Bad rate 16.44% → 9.10%
V2	02	Interpolation stops at the last real observation	Medium	3,849 invented CRWA and 145 invented Capital / RWA values removed
V3	02	Explicit default list read from the register, window anchored on the decree	Low	ind_default_12m 12 → 23 rows
S1	03	C01 and A02 not offered to the model (target leakage)	High	oot AUC 0.899 → 0.864, and the higher figure is the leak
S2	03	Variable direction fitted on train, not assumed	Medium	L01 kept instead of dropped; train 0.784 → 0.814, oot 0.848 → 0.864
S3	03	train and oot split moved to 201703-202212 / 2023	Medium	Out-of-time bad events 100 → 207
S4	03	Binning, VIF, correlation and stepwise fitted on train only	Medium	oot AUC 0.867 → 0.864, no longer optimistic by construction
S5	03	Stepwise scored on roc_auc, not accuracy	Low	Selection path becomes monotone; fitted model unchanged
M1	04	Logit and forest compared on identical WoE inputs and rows	High	A clean Δ AUC of +0.0031 instead of a meaningless 0.862 against 0.816
M2	04	No fillna(0) on raw CAMELS ratios	High	Missing values no longer relabelled as “excellent”
M3	04	SGS series 4189, not 432; no bare except	High	SELIC no longer silently absent from the macro overlay
M4	04	Paired stratified bootstrap, B = 4,000, row-level and clustered	Medium	Every comparison we report now carries a confidence interval
M6	05	Rating scale and PSI flagged as anticipated material	Low	Labelling only, no numeric effect
M7	04	Inner 70/30 split inside train not carried into notebook 04	Low	oot AUC 0.8636 → 0.8652

■ Fifteen recorded deviations, each with the original line, the replacement, the reason and the measured effect.